

School Management System

Functional Requirement Document

Empowering Education Through Smart Digital Management.

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1. Project Overview

The School Management System is a cloud-based SaaS platform designed to automate and simplify the daily operations of educational institutions. The system provides a centralized platform for managing admissions, academics, attendance, examinations, fees, communication, transportation, human resources, and reporting.

Being a SaaS-based solution, the platform allows multiple schools to operate independently under a single Super Admin while maintaining complete data isolation and security for each institution.

2. Project Objective

The objective of this system is to digitize school operations, reduce manual work, improve communication among stakeholders, and provide real-time access to academic and administrative information. The platform helps schools improve operational efficiency while providing a better experience for students, parents, teachers, and administrators.

3. Project Scope

The School Management System covers all major administrative and academic activities required by an educational institution. The platform supports role-based access to ensure every user can perform only the actions assigned to their role.

The system includes modules for student management, academics, attendance, examinations, fee collection, transport, hostel, library, HR, communication, reports, mobile applications, and administration.

4. User Roles

Role	Description
Super Admin	Manages the SaaS platform, subscriptions, schools, and global settings.
School Admin	Manages all academic and administrative operations of an individual school.
Principal	Monitors school performance, academic activities, and staff operations.
Teacher	Manages attendance, assignments, examinations, and student performance.
Student	Accesses attendance, homework, examinations, results, and personal information.
Parent	Monitors their child's academic progress, attendance, fee payments, and school announcements.
Accountant	Handles fee collection, financial records, and payment reports.
Librarian	Manages books, library members, and book transactions.
Transport Manager	Manages vehicles, routes, drivers, and transport assignments.
HR/Admin Staff	Manages employee records, payroll, and administrative operations.

5. System Overview

The application follows a role-based access model where every user logs into the system using secure credentials. After successful authentication, the user is redirected to a personalized dashboard that displays only the modules assigned to their role. All actions performed within the application are recorded to maintain transparency and accountability.

6. Authentication Module

Module Overview

The Authentication Module provides secure access to the application by validating user credentials and controlling access based on user roles and permissions.

6.1 User Login

The system allows users to securely log in using their registered email address, mobile number, employee ID, or student ID along with their password. After successful authentication, users are redirected to their respective dashboards according to their assigned role.

6.2 Forgot Password

Users who forget their password can reset it through OTP verification or email verification. The system validates the user's identity before allowing a new password to be created.

6.3 Role-Based Access Control

Every user is assigned a specific role within the system. Access to modules, menus, reports, and actions is determined according to the permissions associated with that role, ensuring secure and controlled system usage.

6.4 Session Management

The system automatically manages user sessions to improve security. Inactive sessions expire after a predefined duration, and users are required to log in again to continue accessing the application.

6.5 Login Activity

Every successful and unsuccessful login attempt is recorded with the date, time, IP address, and device information. These logs help administrators monitor unauthorized access attempts and maintain system security.

7. Student Management Module

Module Overview

The Student Management Module manages the complete lifecycle of a student, beginning with admission and continuing through academics, attendance, examinations, fee management, and graduation. It acts as the primary repository for all student-related information.

7.1 Student Admission

The system enables schools to manage both online and offline admissions. Applicants can submit admission forms with the required documents, and administrators can verify, approve, or reject applications before generating the student's official record.

7.2 Student Profile

Each student has a centralized profile containing personal information, guardian details, academic records, medical information, uploaded documents, and communication history. Authorized users can update the profile whenever necessary while maintaining an audit history.

7.3 Student ID Generation

The application automatically generates a unique admission number and student ID during the admission process. These identifiers remain unique throughout the student's academic journey.

7.4 Class & Section Assignment

School administrators can assign students to specific classes, sections, and academic sessions. The system ensures that student allocation follows the institution's academic structure.

7.5 Student Promotion

At the end of every academic session, administrators can promote students to the next class while preserving their complete academic history, attendance records, examination results, and fee details.

7.6 Student Transfer

The platform supports transfers between sections, classes, branches, or campuses without losing any academic or administrative records. Previous information remains available for future reference.

7.7 Student Attendance Summary

The student profile displays attendance statistics, leave history, and monthly attendance percentages. Teachers, parents, and students can easily monitor attendance performance through this section.

7.8 Student Document Management

The system stores digital copies of important student documents such as birth certificates, Aadhaar cards, transfer certificates, photographs, medical records, and previous academic certificates. Documents can be securely uploaded, viewed, or updated by authorized users.

7.9 Student Status Management

Administrators can update the status of a student as Active, Inactive, Graduated, Suspended, or Transferred. Status updates help maintain accurate student records throughout the academic lifecycle.

8. Academic Management Module

Module Overview

The Academic Management Module manages the institution's academic structure by organizing academic sessions, classes, sections, subjects, timetables, and curriculum planning. It provides a standardized framework for conducting academic activities efficiently.

8.1 Academic Session

The system allows administrators to create and manage multiple academic years. Each session contains its own students, classes, examinations, fee structures, and academic schedules.

8.2 Class Management

Administrators can create and manage classes according to the institution's academic structure. Classes can be modified, archived, or reused for future academic sessions whenever required.

8.3 Section Management

Each class can contain multiple sections to accommodate different student groups. Administrators can create, edit, merge, or deactivate sections based on the institution's requirements.

8.4 Subject Management

The application allows administrators to define subjects, assign subject codes, categorize theory and practical subjects, and allocate teachers to each subject. Subject configurations can vary according to class and academic session.

8.5 Class Timetable

The timetable module enables administrators to prepare class schedules by assigning teachers, subjects, classrooms, and lecture timings. The system prevents scheduling conflicts and ensures proper timetable management.

8.6 Curriculum Planning

Schools can define yearly syllabi, lesson plans, and academic milestones for every class and subject. Teachers can monitor syllabus completion and update teaching progress throughout the academic year.

8.7 School Calendar

The academic calendar displays holidays, examinations, events, vacations, parent meetings, and other institutional activities. All users receive updated schedules according to their respective roles.

9. Attendance Management Module

Module Overview

The Attendance Management Module helps schools record, monitor, and analyze student and staff attendance. It simplifies daily attendance tracking while providing accurate reports for administrators, teachers, students, and parents.

9.1 Student Attendance

Teachers can mark daily attendance for students based on class and section. Attendance can be recorded as Present, Absent, Late, Half-Day, or Leave according to the school's attendance policy.

9.2 Staff Attendance

The system allows administrators to record attendance for teachers and non-teaching staff. Attendance records are used for payroll processing, leave management, and performance monitoring.

9.3 Attendance Corrections

Authorized users can modify attendance records in case of errors or approved leave requests. Every modification is recorded in the system along with the user who performed the update.

9.4 Leave Management

Students and employees can submit leave applications through the system. The concerned authority can approve or reject the request, and the attendance record is automatically updated after approval.

9.5 Attendance Reports

The system generates daily, weekly, monthly, and yearly attendance reports. Reports can be filtered by class, section, student, teacher, or department for better analysis.

10. Examination & Result Management Module

Module Overview

This module manages the complete examination process, including exam scheduling, marks entry, grading, report cards, and result publication. It provides a structured approach to evaluating student performance.

10.1 Examination Setup

Administrators can create multiple examinations such as Unit Tests, Mid-Term, Final Exams, Practical Exams, or custom assessments. Each examination can be configured independently for different classes.

10.2 Exam Timetable

The system allows schools to prepare and publish examination schedules. Students, parents, and teachers can view the timetable from their respective dashboards.

10.3 Marks Entry

Teachers can enter subject-wise marks or grades for assigned classes. The system validates entered marks and prevents duplicate submissions.

10.4 Grade Management

Schools can configure grading systems such as percentage, GPA, CGPA, or custom grading policies. Grades are calculated automatically based on predefined rules.

10.5 Result Generation

The system automatically generates report cards after marks verification. Results include subject-wise marks, grades, percentage, overall performance, and ranking where applicable.

10.6 Result Publishing

Administrators can publish examination results once they are verified. Students and parents receive notifications when results become available.

10.7 Performance Analysis

The platform provides performance analytics for students, classes, and subjects. These reports help teachers identify strengths, weaknesses, and overall academic progress.

11. Fee Management Module

Module Overview

The Fee Management Module automates fee collection, payment tracking, receipt generation, and financial reporting. It supports both online and offline payment methods while maintaining accurate financial records.

11.1 Fee Structure

Administrators can define fee structures for different classes, academic sessions, and student categories. Multiple fee heads such as tuition, admission, transport, hostel, and examination fees can be configured.

11.2 Fee Collection

The system allows schools to collect payments through cash, cheque, bank transfer, UPI, cards, or integrated online payment gateways. Every payment generates a unique receipt.

11.3 Installment Management

Schools can divide fees into multiple installments based on institutional policies. Students and parents can easily track upcoming payment schedules.

11.4 Discounts & Scholarships

Administrators can provide fee discounts, scholarships, concessions, or special waivers to eligible students. The system automatically adjusts payable amounts.

11.5 Fine Management

Late payment penalties can be configured according to school policies. The system automatically calculates overdue fines whenever applicable.

11.6 Fee Reports

Comprehensive reports display collected fees, pending dues, overdue payments, concessions, refunds, and daily financial summaries for easy accounting.

12. Library Management Module

Module Overview

The Library Management Module helps schools manage books, members, issue records, returns, fines, and inventory through a centralized digital library system.

12.1 Book Management

Librarians can add, update, categorize, and remove books from the library. Each book contains details such as title, author, ISBN, publisher, category, and available copies.

12.2 Book Issue & Return

Books can be issued to students and staff with predefined return dates. The system tracks issued books and updates inventory automatically upon return.

12.3 Fine Calculation

Late return fines are automatically calculated based on the institution's library policy. Librarians can collect or waive fines with proper authorization.

12.4 Library Reports

The system provides reports for issued books, overdue returns, available inventory, damaged books, and library usage statistics.

13. Transport Management Module

Module Overview

The Transport Management Module helps schools manage vehicles, routes, drivers, pickup points, and student transportation efficiently while ensuring student safety.

13.1 Vehicle Management

Administrators can maintain complete records of school buses and vehicles, including registration details, seating capacity, insurance information, and maintenance schedules.

13.2 Route Management

The system allows schools to create transport routes with multiple pickup and drop points. Students can be assigned to routes based on their location.

13.3 Driver Management

Schools can manage driver information, licenses, contact details, and assigned vehicles. Driver records remain available for administrative reference.

13.4 Student Transport Assignment

Students can be assigned to specific vehicles and routes. Parents can view transport details directly from their dashboard.

13.5 Transport Reports

The system generates reports showing route assignments, vehicle occupancy, transport fee collection, and vehicle utilization.

14. Hostel Management Module

Module Overview

The Hostel Management Module helps institutions manage hostel rooms, student accommodation, room allocation, and hostel-related fees through a centralized system.

14.1 Hostel Setup

Administrators can create hostel buildings, floors, room categories, and room capacities according to institutional requirements.

14.2 Room Allocation

Students can be assigned rooms based on availability, gender, room type, or other institutional policies. Room occupancy is updated automatically.

14.3 Hostel Attendance

Hostel wardens can maintain daily attendance records for hostel residents to ensure proper monitoring and security.

14.4 Hostel Fee Management

Hostel charges can be managed separately from academic fees. The system tracks payments, pending dues, and payment history.

14.5 Hostel Reports

Administrators can generate reports related to room occupancy, hostel residents, available rooms, fee collection, and maintenance requirements.

15. Teacher Management Module

Module Overview

The Teacher Management Module helps schools maintain complete teacher records, subject allocations, workload distribution, and academic responsibilities. It centralizes all teacher-related information for efficient administration.

15.1 Teacher Profile

The system stores comprehensive teacher information, including personal details, qualifications, experience, department, contact information, and employment history. Administrators can update records whenever required.

15.2 Subject Allocation

Administrators can assign one or multiple subjects to teachers based on their expertise and academic schedule. Subject assignments can be modified during the academic session if required.

15.3 Class Allocation

Teachers can be assigned to multiple classes and sections for conducting lectures, attendance, examinations, and classroom activities.

15.4 Timetable Management

Teachers can access their daily and weekly teaching schedules from their dashboard. Any timetable updates are reflected automatically.

15.5 Performance Records

The system maintains teacher performance records, including attendance, assigned workload, completed syllabus, and academic activities to assist management during evaluations.

16. Human Resource & Payroll Module

Module Overview

The HR & Payroll Module manages employee records, attendance, leave, salaries, payroll processing, and organizational documentation from a centralized platform.

16.1 Employee Management

Administrators can maintain complete employee records, including teaching and non-teaching staff. Employee profiles include personal information, department, designation, joining date, qualifications, and employment status.

16.2 Leave Management

Employees can apply for different types of leave, including casual leave, sick leave, earned leave, and unpaid leave. Leave requests are processed according to the organization's approval hierarchy.

16.3 Payroll Management

The system calculates employee salaries based on attendance, allowances, deductions, incentives, overtime, and statutory deductions. Salary slips are generated automatically after payroll processing.

16.4 Salary Reports

Administrators can generate monthly salary reports, deduction summaries, payment history, and payroll statements for accounting purposes.

17. Homework & Assignment Module

Module Overview

The Homework & Assignment Module enables teachers to assign academic tasks digitally while allowing students to submit completed work through the application.

17.1 Homework Creation

Teachers can create homework by selecting the class, subject, submission date, and instructions. Supporting documents, images, or reference materials can also be attached.

17.2 Assignment Submission

Students can upload completed assignments before the submission deadline. The system records submission time and maintains submission history.

17.3 Homework Evaluation

Teachers can review submitted assignments, provide marks, comments, and feedback. Students can access teacher remarks directly from their dashboard.

17.4 Homework History

Both teachers and students can view previous assignments, submission records, grades, and pending work for future reference.

18. Communication & Notification Module

Module Overview

The Communication Module provides a centralized communication channel between the school, students, parents, teachers, and staff. It ensures important information reaches the intended users promptly.

18.1 Announcements

School administrators can publish announcements related to examinations, holidays, events, admissions, or emergency notices. Announcements are displayed on user dashboards according to their roles.

18.2 SMS Notifications

The system can send SMS notifications for attendance, fee reminders, examination schedules, admission updates, and important school announcements.

18.3 Email Notifications

Email notifications are used to deliver academic updates, fee receipts, report cards, circulars, and administrative communications to users.

18.4 Push Notifications

Mobile application users receive real-time push notifications for homework, attendance, fee reminders, examination results, and announcements.

18.5 Internal Messaging

Teachers, administrators, parents, and students can communicate through the internal messaging system while maintaining communication history within the platform.

19. Event & Calendar Management Module

Module Overview

The Event Management Module helps schools organize academic and extracurricular activities while providing a centralized calendar for institutional events.

19.1 Event Creation

Administrators can create events such as annual functions, sports competitions, seminars, workshops, cultural programs, and parent-teacher meetings.

19.2 School Calendar

The academic calendar displays holidays, examinations, vacations, admission schedules, and school events for all users.

19.3 Event Participation

Students and teachers can view upcoming events and participate according to event eligibility defined by the administration.

19.4 Event Reports

The system maintains event history, participant lists, attendance records, and event summaries for future reference.

20. Inventory & Asset Management Module

Module Overview

This module helps educational institutions manage school assets, inventory, stationery, laboratory equipment, furniture, and other institutional resources.

20.1 Asset Registration

Administrators can register school assets by recording asset details, purchase information, quantity, supplier details, and asset location.

20.2 Inventory Management

The system tracks available stock, issued items, damaged assets, and inventory movements to maintain accurate stock records.

20.3 Asset Allocation

Assets can be assigned to classrooms, laboratories, departments, or employees while maintaining complete allocation history.

20.4 Inventory Reports

Administrators can generate reports for available stock, low inventory, issued assets, damaged items, and purchase history.

21. Reports & Analytics Module

Module Overview

The Reports & Analytics Module provides comprehensive reports across academic, administrative, financial, and operational activities to support better decision-making.

21.1 Student Reports

The system generates reports related to admissions, attendance, examination performance, fee payments, and student progress.

21.2 Financial Reports

Administrators can access reports showing fee collection, pending payments, salary expenses, income, expenditures, and financial summaries.

21.3 Academic Reports

Academic reports include examination statistics, class performance, subject analysis, teacher workload, and syllabus completion.

21.4 Attendance Reports

Attendance reports provide daily, monthly, and yearly attendance summaries for students, teachers, and staff.

21.5 Custom Reports

Users with appropriate permissions can generate customized reports by selecting filters such as date range, class, department, session, or specific users.

22. Mobile Applications

Module Overview

The School Management System includes dedicated mobile applications to provide convenient access to academic and administrative information for all stakeholders.

22.1 Student Mobile App

Students can access attendance, homework, examination schedules, results, fee details, class timetable, notices, assignments, and academic updates from their mobile devices.

22.2 Parent Mobile App

Parents can monitor their child's attendance, academic performance, fee payments, homework, examination results, transport details, and school announcements in real time.

22.3 Teacher Mobile App

Teachers can mark attendance, manage homework, upload marks, view timetables, communicate with parents, and receive important notifications through the mobile application.

22.4 Admin Mobile App

School administrators can monitor admissions, attendance, fee collection, staff activities, notifications, and important reports directly from their mobile devices.

23. Super Admin Module (SaaS Platform)

Module Overview

The Super Admin is responsible for managing the complete SaaS platform. This role controls multiple schools, subscriptions, licensing, global settings, and monitors overall platform performance.

23.1 School Management

The Super Admin can create, update, activate, deactivate, or remove schools from the platform. Each school operates independently with its own users, data, and configurations.

23.2 Subscription Management

The system supports multiple subscription plans (Monthly, Quarterly, Yearly, or Custom). Super Admin can assign plans, monitor expiry dates, renew subscriptions, and restrict access after expiration.

23.3 License Management

Each school is assigned a license based on its subscription. License limitations may include the number of students, staff members, modules, storage capacity, or application users.

23.4 Module Management

The platform allows the Super Admin to enable or disable specific modules for each school according to the subscribed plan. This provides flexibility for schools to purchase only the features they require.

23.5 Platform Monitoring

The dashboard displays overall statistics such as total schools, active subscriptions, registered students, employees, storage usage, and revenue generated across the platform.

24. School Admin Panel

Module Overview

The School Admin Panel serves as the central control system for managing all academic, financial, and administrative operations of an individual school.

24.1 Dashboard

The dashboard provides a quick overview of admissions, student strength, attendance, examinations, fee collection, employee information, and important announcements through graphical reports.

24.2 User Management

Administrators can create, update, suspend, or deactivate user accounts for teachers, students, parents, accountants, librarians, transport staff, and other employees.

24.3 Permission Management

Role-based permissions ensure that each employee can access only the modules assigned to their responsibilities.

24.4 System Configuration

Administrators can configure academic sessions, grading systems, school timings, attendance rules, fee structures, notification templates, and other institution-specific settings.

24.5 Audit Logs

The system records all important administrative activities, allowing administrators to monitor system changes, user activities, and operational history.

25. Dashboard & Analytics

Module Overview

The Dashboard provides real-time insights into school operations, helping administrators make informed decisions through graphical and statistical reports.

25.1 Dashboard Widgets

The dashboard displays important information such as:

- Total Students
- Total Teachers
- Active Employees
- Today's Attendance
- Fee Collection
- Pending Fees
- Upcoming Examinations
- Recent Admissions
- Notifications

- School Events
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25.2 Graphical Reports

Visual reports help management analyze admissions, attendance trends, financial performance, examination results, and overall institutional growth.

26. Third-Party Integrations

Module Overview

The system supports integration with third-party services to enhance communication, payment processing, security, and overall system functionality.

Supported integrations include:

- Payment Gateway
 - SMS Gateway
 - Email Services
 - WhatsApp Notifications
 - Google Maps
 - Firebase Push Notifications
 - Cloud Storage
 - Video Meeting Platforms
 - Biometric Attendance Devices
 - RFID/Smart Card Systems
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27. Security Module

Module Overview

The Security Module protects sensitive institutional data through secure authentication, authorization, encryption, and continuous activity monitoring.

Security Features

- Role-Based Access Control
 - Password Encryption
 - Multi-Factor Authentication (Optional)
 - Login Activity Monitoring
 - Device Session Management
 - Audit Trail
 - Automatic Logout
 - Data Backup
 - Database Encryption
 - SSL Secure Communication
-

28. Reports Module

Module Overview

The system generates comprehensive reports across every operational module to support management decisions and regulatory compliance.

Available reports include:

- Student Reports
- Admission Reports
- Attendance Reports
- Examination Reports
- Fee Collection Reports
- Salary Reports
- Transport Reports
- Library Reports
- Hostel Reports

- Inventory Reports
- Employee Reports
- Financial Reports
- Custom Reports

Reports can be exported in PDF, Excel, and CSV formats.

29. Non-Functional Requirements

Performance

The application should provide fast response times and efficiently support thousands of concurrent users without performance degradation.

Availability

The platform should maintain high availability with minimal downtime to ensure uninterrupted academic and administrative operations.

Scalability

The architecture should support future expansion, allowing additional schools, users, and modules to be added without affecting existing performance.

Security

All confidential information, including student records, employee data, and financial transactions, should be securely stored and transmitted using industry-standard security practices.

Backup & Recovery

Automatic backups should be scheduled regularly, and the system should support quick recovery to minimize data loss during unexpected failures.

Compatibility

The application should function seamlessly across modern web browsers, Android devices, and iOS devices while maintaining a consistent user experience.

30. Technology Stack

The recommended technology stack for the School Management System includes:

Frontend

- React.js (Web)
- React Native (Android & iOS)

Backend

- Node.js
- Express.js

Database

- MongoDB

Cloud & Storage

- AWS S3 / DigitalOcean Spaces

Authentication

- JWT
- OTP Verification

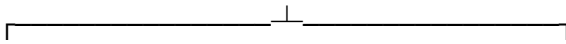
Notifications

- Firebase Cloud Messaging
- SMS Gateway
- Email Service

Payment

- Razorpay
- Stripe (Optional)

Super Admin



Manage Schools

Manage SaaS Settings



Create School



School Admin Login



Configure School Setup

(Academic Session, Classes, Sections, Subjects,

Fee Structure, Roles & Permissions)



Add Staff & Teachers



Student Admission



Assign Class & Section

|



Generate Student ID

|



Daily School Operations

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|— Student Attendance

|— Staff Attendance

|— Homework & Assignments

|— Timetable Management

|— Examination Management

|— Result Generation

|— Fee Collection

|— Library Management

|— Transport Management

|— Hostel Management

|— Events & Announcements

└— Communication

|



Parents & Students Access Information

|



Reports & Analytics



School Administration & Decision Making